

Stack Overflow is a community of 4.7 million programmers, just like you, helping each other. Join them; it only takes a minute:

Sign up



Return type from a Comparator

What does the return value inside the Comparator actually mean?

For example :

```
class TreeSetDemo
{
    public static void main(String arg[])
    {
        TreeSet t=new TreeSet(new MyComparator());
        t.add(new Integer(20));
        t.add(new Integer(10));
        t.add(new Integer(30));
        t.add(new Integer(100));
        System.out.println(t);
    }

    class MyComparator implements Comparator
    {
        public int compare(Object o1, Object o2)
        {
            return 0;
        }
    }
}
```

If the return type is 1 then its actually returning

[20, 10, 30, 100]

If the return type is -1 then its actually returning

[100, 30, 10, 20]

If the return type is 0 then its actually returning

[20]

Please tell me what does this indicate?

java

edited Jun 25 '11 at 15:42



FishBasketGordo
16.6k 3 30 64

asked Jun 25 '11 at 14:54



Preethi Jain
5,249 35 128 216

2 usually 0 indicates == , 1 indicates > and -1 indicates < – [Eng.Fouad](#) Jun 25 '11 at 14:58

2 "1", "-1" and "0" are not types, but values. – [Kerrek SB](#) Jun 25 '11 at 14:59

1 What comparator does is very well laid out in its documentation. Try reading a little before simply asking a question. – [Dunes](#) Jun 25 '11 at 15:06

5 Answers

The return value (not type, the type is int) tells the caller (the thing sorting the data):

```
-1 : o1 < o2
0 : o1 == o2
+1 : o1 > o2
```

If you always return the same value (0, 1, -1) for the comparator, regardless of its inputs, then you're not using it correctly. You need to base the value returned on the values passed in. The idea is that the data structure (or sorter) calls the comparison function any time it needs to order two elements, to find out what order to put them in.

Its worth noting that the positive/negative integer values (-1, +1) don't need to be 1, they can be any positive/negative numbers. It's just common practice to return -1/+1.

answered Jun 25 '11 at 14:58



RHSeeger
10.5k 2 32 36

thank you very very much – [Preethi Jain](#) Jun 25 '11 at 15:00

You are confusing **return-type** and **return-value**. The return-type is `int`. The return value is described in the [documentation](#):

Returns a negative integer, zero, or a positive integer as the first argument is less than, equal to, or greater than the second.

answered Jun 25 '11 at 14:59



[Björn Pollex](#)

45.7k 10 104 196

[Comparator documentation](#) -- Returns: a negative integer, zero, or a positive integer as the first argument is less than, equal to, or greater than the second.

answered Jun 25 '11 at 15:00



[Dunes](#)

13.2k 2 21 40

It's about sort algorithm which needs compare.

Correct:

```
class MyComparator implements Comparator<Integer> {  
    public int compare(Integer o1, Integer o2) {  
        return o1.complare(o2);  
    }  
}
```

answered Jun 25 '11 at 14:59



[卢声远 Shengyuan Lu](#)

13k 11 44 94

Depending on how you want to sort based on this comparator you need to place some logic in the comparator. Your comparator only returns 0 which means equal

```
class MyComparator implements Comparator {  
    public int compare(Object o1, Object o2) {  
        // TODO Auto-generated method stub  
        return o1.compareTo(o2);  
    }  
}
```

edited Jun 25 '11 at 15:13



[Björn Pollex](#)

45.7k 10 104 196

answered Jun 25 '11 at 15:05



[Kevin Bowersox](#)

57.1k 6 65 97

1 I don't think this will compile, because `Object` is not comparable. – [Björn Pollex](#) Jun 25 '11 at 15:14